2.19 Cutting speed, feed and depth of cut

Cutting speed

Speed in general refers to the distance a point moves in a particular period of time. The cutting speed in a drilling operation refers to the peripheral speed of a point on the cutting edge of the drill. It is usually expressed in meters per minute. The cutting speed (v) may be calculated as.

Cutting speed (c.s) $V = \frac{\pi dn}{1000}$ m/min.

Where 'd' is the diameter of the drill in mm,

'n' is the speed of the spindle in rpm and

$\pi = \frac{22}{7}$ (or) 3.14.

Example:

A drill of diameter 20mm makes a hole on a steel part at a cutting speed of 25m/min.

Find out the spindle speed.

Cutting speed (c.s) $V = \frac{\pi dn}{1000}$ m/min

$25 = \pi \times 20xn/1000$

$n = 25x1000/ \pi \times 20 = 398$ rpm.

Spindle speed 'n' = 398 rpm.

Feed

The feed of a drill is the distance. The drill moves into the work at each revolution of the spindle. It is expressed in millimetres. The feed may also be expressed as feed per minute. The feed per minute may be defined as the axial distance moved by the drill into the work per minute. Feed depends upon factors like the material to be drilled. The rigidity of the machine, power, depth of the hole and the type of finish required.

Depth of cut

The depth of cut in drilling is equal to one half of the drill diameter. If 'd' is the diameter of the drill. The depth of cut (t) t=d/2 mm.

2.20 Safety precautions

It is necessary that no damage is done to the operator, the machine parts and the cutting tool. To ensure this the following points are to be remembered.

1. The work should not be held by hand in any case.
2. Proper work holding device should be used to hold the work. If the work is not held properly. The work tends to rotate along with the drill causing damage to the operator, the machine tool and the cutting tool.
3. The shank of the drill and taper hole of the spindle should be cleaned before it is fitted into the spindle.
4. The shank of the drill should confirm with the spindle hole.
5. Cutting speed, feed and depth of cut should be selected according to the prescribed range.
6. Care should be taken to ensure whether the belt and gears are connected properly.
7. Proper safety plates should be installed around rotating parts like belt, drive and gears.
8. The operator should wear safety goggles while operating the drilling machine.
9. The machine should be disconnected from electric terminals when repairs are under taken. In general we should ensure the proper functioning of the machine tools.

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